Sensor + Irrigation Node — Wiring Reference

Board: generic ESP32 DevKit V1 (30-pin). Ground is shared across every circuit below — battery negative, buck converter GND, relay module GND, ESP32 GND, and every sensor's GND all tie to one common ground.

Pins deliberately avoided: GPIO0/2/5/12/15 (boot-mode straps), GPIO1/3 (USB serial, don't repurpose), GPIO6-11 (wired internally to flash, not broken out). GPIO21/22 are reserved free for the I2C RTC module the solar tracker objective will need later.

Circuit 1 — Single soil moisture sensor (test first, before wiring all three)

Resistive sensor module pins: VCC, GND, AO (analog output).

Sensor pin	ESP32 pin
GND	GND
VCC	GPIO25 (digital OUTPUT — pulse HIGH only while reading, to slow corrosion)
AO	GPIO34 (ADC1 input, input-only pin — safe to read)

Test: pulse GPIO25 HIGH, read GPIO34 with analogRead(), set GPIO25 LOW again. Print to serial once a second. Dry air vs. dipped in water should give clearly different raw values — that's your calibration range.

Circuit 2 — All three sensors

Bed	Sensor VCC →	Sensor AO →
Bed 1	GPIO25	GPIO34
Bed 2	GPIO26	GPIO35
Bed 3	GPIO27	GPIO36 (labeled "VP" on most boards)

Circuit 3 — Power chain

```
Solar panel -> Charge controller (PV/Solar terminals)
                      |
               Charge controller (Battery terminals)
                      |
                  12V Battery
                      |
              [ INLINE FUSE ]  <- right after battery +, before any
branch
                      |
             Fused positive bus
              |                          |
     Buck converter IN+       Relay module power (COM side, Circuit
4)
     Buck converter IN- -> GND
```

```
        |
Buck converter OUT+ -> ESP32 5V/VIN
Buck converter OUT- -> ESP32 GND
```

Adjust the buck converter's trimmer pot to a steady 5.0V (check with a multimeter) BEFORE connecting it to the ESP32. Pump and valves run on raw 12V from the fused bus, switched through the relay — never route them through the buck converter or the ESP32's 5V rail.

Circuit 4 — Relay module (4-channel: pump + 3 valves)

Relay channel	ESP32 pin	Switches
IN1	GPIO18	Pump
IN2	GPIO19	Valve 1 (Bed 1)
IN3	GPIO4	Valve 2 (Bed 2)
IN4	GPIO23	Valve 3 (Bed 3)

Relay VCC -> same 5V as ESP32 (buck converter output). Relay GND -> common ground.

Switched side, per channel: fused 12V positive bus -> relay COM; relay NO -> pump's or valve's positive lead; pump's/valve's negative lead -> common ground. The relay only ever switches the positive wire.

Gotcha: most cheap relay boards are active-LOW (relay energizes on LOW, not HIGH). Test empirically if there's no datasheet/silkscreen confirming polarity.

Test: cycle each relay channel on/off one at a time with no sensor logic involved yet — confirm each one clicks/buzzes/runs on cue before combining with sensing.

Full pin map

Function	GPIO
Bed 1 sensor VCC pulse	25
Bed 1 sensor AO (ADC)	34
Bed 2 sensor VCC pulse	26
Bed 2 sensor AO (ADC)	35
Bed 3 sensor VCC pulse	27
Bed 3 sensor AO (ADC)	36 (VP)
Pump relay	18
Valve 1 relay	19
Valve 2 relay	4
Valve 3 relay	23
Reserved (I2C RTC, later)	21 (SDA), 22 (SCL)